

A translation approach to portable ontology specifications

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To support the sharing and reuse of formally represented knowledge among AI systems, it is useful to define the common vocabulary in which shared knowledge is represented. A specification of a representational vocabulary for a shared domain of discourse—definitions of classes, relations, functions, and other objects—is called an ontology. This paper describes a mechanism for defining ontologies that are portable over representation systems. Definitions written in a standard format for predicate calculus are translated by a system called Ontolingua into specialized representations, including frame-based systems as well as relational languages. This allows researchers to share and reuse ontologies, while retaining the computational benefits of specialized implementations.

We discuss how the translation approach to portability addresses several technical problems. One problem is how to accommodate the stylistic and organizational differences among representations while preserving declarative content. Another is how to translate from a very expressive language into restricted languages, remaining system-independent while preserving the computational efficiency of implemented systems. We describe how these problems are addressed by basing Ontolingua itself on an ontology of domain-independent, representational idioms.

1. Introduction

A body of formally represented knowledge is based on a *conceptualization*: the objects, concepts, and other entities that are presumed to exist in some area of interest and the relationships that hold among them (Genesereth & Nilsson, 1987). A conceptualization is an abstract, simplified view of the world that we wish to represent for some purpose. Every knowledge base, knowledge-based system, or knowledge-level agent is committed to some conceptualization, explicitly or implicitly.

An *ontology* is an explicit specification of a conceptualization. The term is borrowed from philosophy, where an ontology is a systematic account of Existence. For knowledge-based systems, what “exists” is exactly that which can be represented. When the knowledge of a domain is represented in a declarative formalism, the set of objects that can be represented is called the universe of discourse. This set of objects, and the formalized relationships among them, are reflected in the representational *vocabulary* with which a knowledge-based program represents knowledge. Thus, we can describe the ontology of a program by defining a set of representational terms. In such an ontology, definitions associate the names of entities in the universe of discourse (e.g. classes, relations, functions, or other objects) with human-readable text describing what the names are meant to denote, and formal axioms that constrain the interpretation and well-formed use of these terms.

This paper addresses the problem of portability for ontologies. Portability is a problem because the parties to a common ontology may use different representation languages and systems. Ideally, shared terms should be defined at the knowledge level, independent of specific representation languages. Of course, definitions need to be couched in *some* common formalism if they are to be shareable by knowledge-based applications. However, it is not realistic or desirable to require that those applications be *implemented* in a common representation language or system. This is because different applications require different kinds of reasoning services, and special-purpose languages to support them. Thus, the portability problem for ontologies is to support common ontologies over multiple representation systems.

We describe a translation approach to the portability problem for ontologies. In a translation approach, ontologies are specified in a standard, system-independent form and translated into specific representation languages. In Section 3 we will describe Ontolingua, an implemented system for translating ontologies from a declarative, predicate-calculus language into a variety of representation systems. In Section 4 we discuss the strengths and limitations of the approach. But first, we define the notion of a shared ontology and describe the role of ontologies for sharing knowledge among AI systems.

2. Ontologies and knowledge sharing

Knowledge-based systems and services are expensive to build, test, and maintain. A software engineering methodology based on formal specifications of shared resources, reusable components and standard services is needed. We believe that specifications of shared vocabulary can play an important role in such a methodology.

Several technical problems stand in the way of shared, reusable knowledge-based software. Like conventional applications, knowledge-based systems are based on heterogeneous hardware platforms, programming languages, and network protocols. However, knowledge-based systems pose special requirements for interoperability. Such systems *operate on and communicate using statements in a formal knowledge representation*. They ask queries and give answers. They take "background knowledge" as an input. And as agents in a distributed AI environment, they negotiate and exchange knowledge. For such knowledge-level communication, we need conventions at three levels: representation language format, agent communication protocol, and specification of the content of shared knowledge. Proposals for standard knowledge representation formats (Morik, Causse & Boswell, 1991; Spiby, 1991; Fulton, 1992; Genesereth & Fikes, 1992) and agent communication languages (Finin *et al.*, 1992) are independent of the content of knowledge being exchanged or communicated. Ontologies can be used for conventions of the third kind: *content-specific* specifications (Gruber, 1991).

Current research is exploring the use of formal ontologies for specifying content-specific agreements for a variety of knowledge-sharing activities (Fikes *et al.* 1991; Neches *et al.*, 1991; Genesereth, 1992; Gruber, Tenenbaum & Weber, 1992; Patil *et al.*, 1992; Walther, Eriksson & Musen, 1992; Cutkosky *et al.*, 1993). A long-term objective of such work is to enable libraries of reusable knowledge components and knowledge-based services that can be invoked over networks.